PHYSICS EXAMINATION BANK

Compilation of Previous Year Papers (Set A and Set B) — Proper English Edition

Core Curriculum Modules: Kinematics, Laws of Motion, Work Energy Power, Gravitation, Electromagnetism, Optics & Waves

Evaluation Level: Advanced Calculus-Based Undergraduate Reference Matrix

Inclusions: Comprehensive Rigorous Step-by-Step Mathematical Analytical Formulations

Verification Code: ENG-PHYS-77X-2026

PREVIOUS YEAR QUESTION PAPER — SET A

Time Allowed: 3 Hours | Maximum Marks: 100 | Instructions: All questions are compulsory. Show complete vector calculus configurations where necessary.

1. Kinematics & Laws of Motion

Q1. A particle moves along the x-axis such that its position as a function of time *t* is given by the equation *x(t) = lpha t^3 - eta t^2*. Determine the precise time instant at which the instantaneous acceleration of the particle becomes identically zero.

A) ***t = eta / (3 lpha)*** [CORRECT]

B) ***t = 2 eta / (3 lpha)***

C) ***t = eta / lpha***

D) ***t = 0***

> **Analytical Rationale**First, compute the instantaneous velocity by taking the first time derivative of position: ***v(t) = dx/dt = 3 lpha t^2 - 2 eta t*** .
> Next, differentiate the velocity function with respect to time to extract the acceleration profile: ***a(t) = dv/dt = 6 lpha t - 2 eta*** .
> Setting the acceleration function to zero to satisfy the equilibrium condition yields: ***6 lpha t - 2 eta = 0 \implies t = 2 eta / 6 lpha = eta / 3 lpha*** .

Q2. A smooth wedge of mass *M* featuring an inclination angle *heta* supports a block of mass *m* on its slanted surface. Find the constant horizontal acceleration *a* that must be applied to the wedge so that the block remains completely stationary relative to the wedge.

A) ***a = g \cdot sin heta***

B) ***a = g \cdot cos heta***

C) ***a = g \cdot tan heta*** [CORRECT]

D) ***a = g / tan heta***

> **Analytical Rationale:** In the non-inertial accelerating reference frame of the wedge, a fictitious pseudo-force of magnitude ***ma*** acts horizontally in the opposite direction on the block.
> Resolving the forces parallel to the inclined plane surface establishes the balance equation: ***ma \cdot cos heta = mg \cdot sin heta \implies a = g \cdot rac{sin heta}{cos heta} = g \cdot tan heta***.

---

2. Work, Energy & Gravitation Fields

Q3. A one-dimensional variable force field defined by the function *F(x) = kx^2* acts on a mass, displacing it from the spatial origin *x = 0* to a terminal position *x = L*. Calculate the total mechanical work performed by this field.

A) *W = kL^3 / 3* [CORRECT]

B) *W = kL^2 / 2*

C) *W = kL^3*

D) *W = 3kL^3*

> **Analytical Rationale**For non-uniform or variable force configurations, work done is calculated via definite integration over the displacement boundaries: *W = \int_0^L F(x)dx = \int_0^L kx^2 dx = k \left[rac{x^3}{3} ight]_0^L = rac{kL^3}{3}* .

Q4. At what precise altitude height *h* above the boundary surface of the Earth does the local acceleration due to gravity drop to exactly half (*1/2*) of its standard sea-level surface value *g*? (Let *R* denote the planetary radius).

A) *h = (\sqrt{2} - 1)R* [CORRECT]

B) *h = R / 2*

C) *h = \sqrt{2}R*

D) *h = (\sqrt{2} + 1)R*

> **Analytical Rationale**The gravitational field intensity variation with altitude follows the exact inverse-square formulation: *g_h = rac{g}{(1 + h/R)^2}* .
> Substituting the given condition *g_h = rac{g}{2}* yields: *rac{g}{2} = rac{g}{(1 + h/R)^2} \implies (1 + h/R)^2 = 2 \implies 1 + h/R = \sqrt{2} \implies h = (\sqrt{2}-1)R* .

---

3. Electromagnetism & Wave Physical Optics

Q5. According to the foundational modifications introduced by James Clerk Maxwell to unify electrodynamics, what is the exact mathematical expression for the Displacement Current (*I_d*)?

A) *I_d = \epsilon_o \cdot (d\Phi_E / dt)* [CORRECT]

B) *I_d = \mu_o \cdot (d\Phi_B / dt)*

C) *I_d = \oint B \cdot dl*

D) *I_d = 0*

> **Analytical Rationale:** To preserve charge conservation and satisfy the continuity equation within capacitive networks or regions with time-varying fields, Maxwell introduced displacement current as proportional to the time rate of change of the electric flux matrix: *I_d = \epsilon_o rac{d\Phi_E}{dt}* .

Q6. A standard Young's Double Slit Experiment (YDSE) apparatus characterized by constant fringe width *eta* in air is fully immersed into a liquid medium possessing a refractive index of *\mu = 4/3*. What is the new fringe width of the observed interference pattern?

A) *eta' = 3 eta / 4* [CORRECT]

B) *eta' = 4 eta / 3*

C) *eta' = eta*

D) The pattern is completely obliterated

> **Analytical Rationale:** When transitioning into an optically denser medium, the wave frequency remains invariant, but the wavelength contracts according to the refractive matrix: *\lambda' = \lambda / \mu*.
> Since the linear fringe spacing follows the ratio *eta = rac{\lambda D}{d}*, the new fringe scaling evaluates directly to: *eta' = rac{\lambda' D}{d} = rac{\lambda D}{\mu d} = rac{ eta}{\mu} = rac{ eta}{4/3} = rac{3 eta}{4}*.

PREVIOUS YEAR QUESTION PAPER — SET B

Time Allowed: 3 Hours | Maximum Marks: 100 | Instructions: The questions below are entirely distinct in numerical and structural formulations from Set A.

1. Kinematics & Laws of Motion

Q1. A projectile is launched from flat horizontal ground at an angle *heta* with respect to the horizontal. If the maximum vertical height attained by the projectile is mathematically equal to its total horizontal range (*H* = *R*), find the exact angle of projection.

A) ***heta = tan^{-1}(1) = 45^\circ***

B) ***heta = tan^{-1}(4)*** [CORRECT]

C) ***heta = tan^{-1}(2)*** D) ***heta = 60^\circ***

> **Analytical Rationale:** Equate the structural equations for Maximum Height and Horizontal Range:
> ***H = rac{u^2 sin^2 heta}{2g}*** and ***R = rac{u^2 sin(2 heta)}{g} = rac{2u^2 sin heta cos heta}{g}***.
> Setting ***H = R \implies rac{u^2 sin^2 heta}{2g} = rac{2u^2 sin heta cos heta}{g} \implies rac{sin heta}{2} = 2cos heta \implies rac{sin heta}{cos heta} = 4 \implies tan heta = 4 \implies heta = tan^{-1}(4)***.

Q2. Two masses *m_1 = 4 ext{ kg}* and *m_2 = 6 ext{ kg}* are connected via an inextensible massless string looped over a frictionless pulley in a standard Atwood machine configuration. Determine the ideal linear acceleration *a* of the system when released from rest. (Take acceleration due to gravity as *g*).

A) ***a = g / 5*** [CORRECT]

B) ***a = g / 10***

C) ***a = g***

D) ***a = 2g / 3***

> **Analytical Rationale**Setting up the net moving forces balance for an Atwood system yields the global acceleration equation:
> ***a = rac{(m_2 - m_1)g}{m_1 + m_2}*** .
> Substituting the given scalar parameters: ***a = rac{(6 - 4)g}{4 + 6} = rac{2g}{10} = rac{g}{5}*** .

---

2. Work, Energy & Gravitation Fields

Q3. Within a conservative vector field, what is the precise mathematical multi-dimensional calculus differential operator linking the force vector field *ec{F}* to the scalar potential energy distribution function *U*?

A) *ec{F} = ec{ abla}U*

B) *ec{F} = - ec{ abla}U* [CORRECT]

C) *U = ec{ abla} \cdot ec{F}*

D) *ec{F} = \partial^2 U / \partial x^2*

> **Analytical Rationale**By definition, a conservative force field is mathematically defined as the negative gradient of its corresponding scalar potential energy function. This ensures path independence of work done: *ec{F} = - ec{ abla}U = -\left(rac{\partial U}{\partial x}\hat{i} + rac{\partial U}{\partial y}\hat{j} + rac{\partial U}{\partial z} \hat{k} ight)* .

Q4. Kepler's Second Law of Planetary Motion states that the areal velocity of a planet tracking an elliptical orbit around a central star remains constant (*dA/dt = ext{constant}*). This empirical deduction is a direct consequence of which fundamental conservation vector framework?

A) Conservation of Linear Momentum

B) Conservation of Total Mechanical Energy

C) Conservation of Angular Momentum [CORRECT]

D) Conservation of Relativistic Mass-Energy

> **Analytical Rationale:** Gravitational force fields are central forces, meaning the force line passes exactly through the coordinate system origin (the star). Consequently, the net external twisting torque acting on the planet is strictly zero: *ec{ au} = ec{r} imes ec{F} = 0*. Because torque is zero, the orbital angular momentum vector *ec{L}* is completely conserved, directly yielding a constant rate of sweeping area: *rac{dA}{dt} = rac{L}{2m} = ext{constant}*.

3. Electrodynamics Matrix & Wave Physical Optics

Q5. Which of Maxwell's field equations explicitly establishes the physical law that isolated single magnetic monopoles cannot exist in classical electrodynamics?

A) *ec{ abla} \cdot ec{E} = ho / \epsilon_o*

B) ***ec{ abla} imes ec{B} = \mu_o ec{J}***

C) ***ec{ abla} \cdot ec{B} = 0*** [CORRECT]

D) ***ec{ abla} imes ec{E} = -\partial ec{B}/\partial t***

> **Analytical Rationale:** Gauss's Law for Magnetism states that the divergence of the magnetic field vector field is identically zero everywhere (***ec{ abla} \cdot ec{B} = 0***). This means that magnetic field flux lines always form completely continuous, endless closed loops, ensuring that net flux through any closed Gaussian matrix surface remains zero, confirming the non-existence of separate magnetic point charge monopoles.

Q6. According to Malus's Linear Polarization Law, if a plane-polarized light beam of baseline intensity *I_o* passes through an optical analyzer whose transmission axis is oriented at an angle of *heta = 60^\circ* relative to the polarizer axis, what is the output transmitted intensity *I*?

A) ***I = I_o / 2***

B) ***I = I_o / 4*** [CORRECT]

C) ***I = 3I_o / 4***

D) ***I = 0***

> **Analytical Rationale:** Applying the exact trigonometric formulation for Malus's law yields: ***I = I_o \cdot cos^2 heta*** .
> Substituting the given angular displacement parameter: ***I = I_o \cdot \left(cos(60^\circ) ight)^2 = I_o \cdot \left(rac{1}{2} ight)^2 = rac{I_o}{4}*** .
